



Robust Bayesian high-dimensional variable selection and inference with the horseshoe family of priors

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ABSTRACT

Frequentist robust variable selection has been extensively investigated in high-dimensional regression. Despite success, developing the corresponding statistical inference procedures remains a challenging task. Recently, tackling this challenge from a Bayesian perspective has received much attention. In literature, the two-group spike-and-slab priors that can induce exact sparsity have been demonstrated to yield valid inference in robust sparse linear models. Nevertheless, another important category of sparse priors, the horseshoe family of priors, including horseshoe, horseshoe+, and regularized horseshoe priors, has not yet been examined in robust high-dimensional regression by far. Their performance in variable selection and especially statistical inference in the presence of heavy-tailed model errors is not well understood. To address this gap, robust Bayesian hierarchical models incorporating the horseshoe family of priors are developed, together with an efficient Gibbs sampling scheme for posterior computation. Numerical studies demonstrate that, compared with competing methods employing alternative sampling strategies such as slice sampling, the proposed methods lead to superior performance in variable selection, Bayesian estimation and statistical inference. In particular, even without imposing exact sparsity, the one-group horseshoe priors can still yield valid Bayesian credible intervals under robust high-dimensional linear regression models. Applications to real data further demonstrate the advantages of the proposed methods over competitors.

1. Introduction

In bioinformatics studies, robust variable selection has been extensively developed and conducted to identify important genomics features associated with complex disease traits such as cancer outcomes (Wu and Ma, 2015). The robustness of variable selection procedures lies in the robust loss functions, such as Huber loss, quantile check loss and rank based loss, among many others, that can downweigh the influence of outliers in obtaining shrinkage estimators compared to the non-robust penalized least square loss. In high-dimensional models, statistical inference plays a critical role in ensuring the superiority of shrinkage estimators and identification results (Dezeure et al., 2015). Nevertheless, frequentist robust variable selection methods with valid finite-sample inference remain challenging to establish and are considerably less developed than their non-robust counterparts (Fan et al., 2024).

The limitation of frequentist robust variable selection methods in statistical inference has motivated us to tackle the problem from a Bayesian perspective. Fully Bayesian analysis is well acknowledged to provide uncertainty quantification measures to enable

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Table 1

An incomplete list of baseline Bayesian sparse models with horseshoe (HS) priors.

Reference	Likelihood (robustness)	Prior	Inference	Implementation
Carvalho et al. (2010)	Normal (No)	HS	Not examined	MCMC (R)
Bhadra et al. (2017)	Normal (No)	HS +	Not examined	MCMC (R)
Piironen and Vehtari (2017)	Normal (No)	Regularized HS	Not examined	NUTS (stan)
van der Pas et al. (2017)	Normal (No)	HS	Yes (low-dim)	MCMC (R) van der Pas et al. (2022)
Kohns and Szendrei (2024)	ALD (Yes)	HS	Not examined	Slice sampling (R)
Fan et al. (2026)	ALD (Yes)	HS, HS +, Regularized HS	Yes (high-dim)	Gibbs sampling (C++)

exact statistical inference. Even on finite samples, as long as the full posterior distribution is accessible via MCMC, marginal credible intervals can be readily computed for inference. Sparse robust Bayesian hierarchical models are built on two components: (1) a robust likelihood and (2) shrinkage priors. Although a vast amount of shrinkage priors have been proposed in literature ([O’hara and Sillanpää, 2009](#)), the spike-and-slab priors ([Mitchell and Beauchamp, 1988](#); [George and McCulloch, 1993](#); [Bai et al., 2021](#)) and horseshoe priors ([Carvalho et al., 2010](#); [Polson and Scott, 2010](#); [Bhadra et al., 2019](#)) are among the most widely studied ones. The spike-and-slab prior is a two-group model that treats zero and non-zero regression coefficients differently *a priori*, typically with a point mass at zero and a continuous density away from zero. In contrast, horseshoe priors form a one-group model that globally shrinks all regression coefficients toward zero, while allowing some to escape away from zero through the “shrink globally, act locally” mechanism ([Polson and Scott, 2010](#)).

Recent studies have shown that robust Bayesian LASSO with spike-and-slab priors (RBLSS) produces marginal Bayesian credible intervals with nominal coverage probabilities even under heavy-tailed model errors ([Ren et al., 2023](#); [Fan et al., 2024](#)). Therefore, a natural next question is whether the one-group horseshoe priors can yield superior shrinkage estimation, variable selection, and, in particular, valid statistical inference under robust hierarchical models. The main empirical argument from [Fan et al. \(2024\)](#) in support of RBLSS’s valid inference procedures is that the spike-and-slab priors induce exact sparsity, which aligns with the sparsity assumption in high-dimensional regression. Although one-group horseshoe priors do not yield exact zero posterior estimates directly, published literature has shown that they can still provide valid inference under sparse normal mean models ([van der Pas et al., 2017](#)). This suggests that their favorable inferential properties may also extend to robust settings.

In this article, we address the above question by developing robust hierarchical models utilizing the horseshoe prior family, including horseshoe, horseshoe+ and regularized horseshoe priors ([Carvalho et al., 2010](#); [Bhadra et al., 2017](#); [Piironen and Vehtari, 2017](#)) for Bayesian shrinkage estimation and inference. Robust likelihood functions, the other critical component (in addition to shrinkage priors) to formulate robust sparse Bayesian hierarchical models, are usually developed based on (1) heavy-tailed distributions, (2) Bayesian nonparametrics methodology such as Dirichlet process mixture models ([Kottas and Krnjajić, 2009](#)), and (3) the robust generalized framework ([Bissiri et al., 2016](#)). In our study, we adopt the Laplace likelihood to impose robustness for important computational considerations. As the Laplace distribution can be expressed as a conditional scaled normal distribution ([Kozumi and Kobayashi, 2011](#)), fast Gibbs sampling with horseshoe priors can be readily extended to robust hierarchical models ([Wand et al., 2011](#); [Makalic and Schmidt, 2015](#)).

In fact, the benefits arising from the connection between a normal likelihood and an (asymmetric) Laplace likelihood for robust analysis have rarely been acknowledged, or at least explicitly discussed, in the published literature. We show two pieces of evidence here. First, in terms of estimation, [Liu et al. \(2024\)](#) show that updates of sparse regression coefficients follow a soft-thresholding rule under robust spike-and-slab quantile LASSO utilizing a Laplacian working likelihood. This phenomenon is rarely observed in frequentist robust penalization due to the striking contrast between the non-differentiable L1 loss and the smooth least square loss ([Ročková and George, 2018](#)). Second, in terms of inference, Bayesian LASSO with spike-and-slab priors (BLSS) yields valid credible intervals under normal model errors ([Fan et al., 2024](#)). It is not surprising that RBLSS also performs well, given the superior performance in inference already demonstrated by its non-robust counterpart BLSS. Therefore, we adopt the Laplace working likelihood to leverage the strengths of horseshoe priors developed under the sparse normal mean model for a more robust setting.

We explore the relevant published literature, as summarized in [Table 1](#), to further demonstrate how our study differs from existing work. [Table 1](#) is consistent with the survey by [Bhadra et al. \(2019\)](#) in that horseshoe prior sparse regressions are mostly focused on non-robust, normal settings and ignore heavy-tailed model errors. Although [van der Pas et al. \(2017\)](#) have established uncertainty quantification properties of horseshoe priors under the sparse normal mean model, these results are limited to low-dimensional settings with Gaussian errors. In addition, [Kohns and Szendrei \(2024\)](#) appears to be the only published study on robust horseshoe regression. Primarily focused on forecasting, it does not address high-dimensional variable selection and inference questions. Furthermore, our numerical studies show that the slice sampling proposed in [Kohns and Szendrei \(2024\)](#) leads to much slower computational speed and inferior performance in variable selection, shrinkage estimation and especially statistical inference compared to our methods.

In summary, the major contributions of our study are as follows:

- First, unlike the slice sampling adopted for horseshoe prior quantile regression ([Kohns and Szendrei, 2024](#)), we have developed Gibbs samplers for robust regression with the family of horseshoe priors, including horseshoe, horseshoe+ and regularized horseshoe priors, by utilizing the auxiliary mixture sampling strategy ([Wand et al., 2011](#); [Makalic and Schmidt, 2015](#)). We present detailed Gibbs samplers that have not been published in any existing work before.
- Second, we demonstrate, for the first time, the variable selection performance of robust Bayesian regression with representative horseshoe priors. Different from the forecasting-oriented horseshoe prior quantile regression ([Kohns and Szendrei, 2024](#)) where

variable selection has not been investigated, we have conducted comprehensive simulations to assess the performance of proposed methods, indicating that the horseshoe family of priors yields a distinctive variable selection pattern under aggressive shrinkage, even though it does not induce exact sparsity.

- Third, among the first time, we report high-dimensional inference results for robust horseshoe regression to address the question that whether the one-group robust horseshoe models can yield valid inference results when “exact sparsity” does not hold. Our numerical analysis indicates that horseshoe type of priors, especially horseshoe and horseshoe+ priors, can produce promising inference results under robust sparse models, facilitating readily cross-comparison with existing high-dimensional inference results (Dezeure et al., 2015; Fan et al., 2024).

- Fourth, we have investigated the connection between posterior sampling schemes and robust high-dimensional inference, which has been largely overlooked in published studies. We show that, compared to robust horseshoe models with alternative sampling schemes (Kohns and Szendrei, 2024; Makalic and Schmidt, 2016), the proposed Gibbs sampling leads to superior performance especially in terms of accurate statistical inference in sparse robust models.

To ensure fast computation and reproducible research, we implement all the proposed and non-robust alternatives in C++ via the *pqrBayes* package (version 1.2.0), which is available on CRAN: <https://cran.r-project.org/package=pqrBayes>. We conclude the paper with a case study on high dimensional eQTL data.

2. Statistical methods

2.1. Robust Bayesian regression

Consider a linear model of the form:

$$y_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \quad \text{for } i = 1, \dots, n, \quad (1)$$

where $\{y_i\}_{i=1}^n$ denotes a scalar response variable, $\{\mathbf{x}_i\}_{i=1}^n$ represents a known $p \times 1$ vector of covariates and random errors ϵ_i are assumed to be *i.i.d.* The intercept is denoted as β_0 . Estimated regression coefficient $\hat{\boldsymbol{\beta}}$ can be obtained by minimizing the following robust least absolute deviation (LAD) loss:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n |y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta}|,$$

which is resistant to outliers due to the L1 form of the LAD loss. More broadly, the LAD loss is a special case of the check loss function in quantile regression at the 50%th quantile level. Model fitting procedures to obtain the corresponding minimizer have been widely discussed in frequentist literature. In our study, to motivate robust Bayesian estimation and inference, we consider the Bayesian median regression corresponding to model (1) by assuming that the error ϵ_i follow *i.i.d.* Laplace distribution, or $f(\epsilon_i|\tau) = \frac{\tau}{4} \exp\{-\frac{\tau}{2}|\epsilon_i|\}$, $i = 1, \dots, n$, where $(\frac{\tau}{2})^{-1}$ is the scale parameter of the Laplace distribution. This corresponds to the asymmetric Laplace distribution parametrized at the 50% quantile level, which is symmetric and commonly used in Bayesian quantile regression (Yu and Moyeed, 2001). In our study, we focus on the median (50% quantile level), and therefore no asymmetry is being modeled. Under the aforementioned probabilistic assumption on the model errors, the conditional distribution of y_i , given the regression coefficients specified in model (1), can be formally expressed as

$$f(y_i | \beta_0, \mathbf{x}_i^\top \boldsymbol{\beta}, \tau) = \frac{\tau}{4} \exp\left(-\frac{\tau}{2}|y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta}|\right). \quad (2)$$

To facilitate the setup of a robust Bayesian hierarchical model, we utilize an alternative representation of the Laplace distribution, by expressing it as a mixture of an exponential distribution and a scaled normal distribution (Kozumi and Kobayashi, 2011). In particular, by setting $\xi = \sqrt{8}$, the error term ϵ_i , which follows a Laplace distribution with scale parameter $2\tau^{-1}$, admits the following hierarchical representation: $\epsilon_i = \tau^{-1}\xi\sqrt{v_i}z_i$, where $v_i \sim \text{Exp}(1)$ and $z_i \sim N(0, 1)$. Accordingly, model (1) can be reformulated as:

$$y_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \tau^{-1}\xi\sqrt{v_i}z_i, \quad \text{for } i = 1, \dots, n,$$

Let $\tilde{v}_i = \tau^{-1}v_i$, then $\tilde{v}_i \stackrel{\text{ind}}{\sim} \text{Exp}(\tau^{-1})$. The Bayesian hierarchical model has the following representation:

$$\begin{aligned} y_i &= \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \tau^{-1/2}\xi\sqrt{\tilde{v}_i}z_i, \\ \tilde{v}_i | \tau &\stackrel{\text{ind}}{\sim} \text{Exp}(\tau^{-1}), \\ z_i &\stackrel{\text{ind}}{\sim} N(0, 1). \end{aligned}$$

Remark: As pointed out by Yang et al. (2016), the above reparameterization of a Laplace distribution leads to a working likelihood that matches its maximum likelihood estimation to the minimization of LAD loss under the model (1). In addition to its widespread use in Bayesian regularized quantile regression, this formulation has recently been noted to bridge frequentist and Bayesian paradigms of robust variable selection (Liu et al., 2024).

2.2. Robust Bayesian regression with the horseshoe family of priors

With high-dimensional regression, sparse priors play a critical role in ensuring superior performance in variable selection, shrinkage estimation and especially statistical inference. In fact, the limitation of adopting the aforementioned likelihood has been pointed out in [Yang et al. \(2016\)](#), showing that with the (asymmetric) Laplace based working likelihood, a posterior variance adjustment step is needed to retain valid inference in low-dimensional settings. Although the variance correction is not available when $p > n$, it has been demonstrated that if the sparse prior is chosen carefully, such as the point-mass spike-and-slab priors, robust Bayesian variable selection can yield valid inference procedures that can be verified on finite samples ([Fan et al., 2024](#); [Ren et al., 2023](#)). An important question we attempt to address is whether the horseshoe family of priors can achieve comparable or even superior performance, particularly in inference under robust high-dimensional linear models.

2.2.1. Robust horseshoe prior regression

[Carvalho et al. \(2010\)](#) and [Polson and Scott \(2010\)](#) have developed the horseshoe prior for continuous shrinkage and sparse signal recovery. The hierarchical form of the horseshoe prior under the robust likelihood specified in [Section 2.1](#) can be expressed as follows.

$$\begin{aligned}\beta_j | s_j^2, \lambda^2 &\sim N(0, s_j^2 \lambda^2), \quad j = 1, \dots, p, \\ s_j &\sim C^+(0, 1), \quad j = 1, \dots, p, \\ \lambda &\sim C^+(0, 1),\end{aligned}\tag{3}$$

which slightly differs from the formulation under normal mean models since the variance parameter that exists in those models is no longer applicable here. The hierarchical model specifically utilizes half-Cauchy distributions for both the local scale parameter s_j and the global scale parameter λ , characterized by the density function $p(x) = \frac{2}{\pi(1+x^2)}$, $x > 0$.

The horseshoe prior has a global-local shrinkage mechanism in that the global shrinkage parameter λ applies uniform shrinkage over all β_j to push regression coefficients toward zero globally. Meanwhile, the heavy-tailed half-Cauchy priors on the local scales s_j allow individual coefficients to resist the shrinkage and remain large when necessary. By adjusting λ , one can control the level of sparsity in the model: a large value of λ leads to weak shrinkage and diffuse priors for all variables, whereas letting $\lambda \rightarrow 0$ forces all coefficients β_j toward zero.

To better understand how the shrinkage is imposed under the half-Cauchy priors which concentrate substantial mass near zero with heavy tails, we establish the following two propositions on the shrinkage factor κ_j from robust horseshoe regression models. The details of derivations can be found in the Appendix.

Proposition 1. Consider the full conditional posterior distribution of β_j given the data and other parameters: $\beta_j | \cdot \sim N(\mu_{\beta_j}, \sigma_{\beta_j}^2)$, then the posterior mean can be expressed as

$$\mu_{\beta_j} = (1 - \kappa_j) \hat{\beta}_j,$$

where the shrinkage coefficient κ_j is defined as $\kappa_j = \left(1 + \lambda^2 s_j^2 a_j\right)^{-1} \in (0, 1)$, with $a_j = \sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1} \xi^2 \tilde{v}_i}$ and $\hat{\beta}_j = a_j^{-1} \sum_{i=1}^n \frac{x_{ij}(y_i - \beta_0 - \sum_{k \neq j} x_{ik} \beta_k)}{\tau^{-1} \xi^2 \tilde{v}_i}$.

This decomposition illustrates how the shrinkage factor κ_j governs the degree to which $\hat{\beta}_j$ is pulled towards zero. Specifically, when the scale parameters λ or s_j approach 0, the shrinkage coefficient κ_j goes to 1. Therefore, the posterior mean $\mu_{\beta_j} \rightarrow 0$ to induce almost zero regression coefficient. On the other hand, signals are retained, i.e., $\mu_{\beta_j} \rightarrow \hat{\beta}_j$ as $\lambda \rightarrow \infty$ or $s_j \rightarrow \infty$.

Proposition 2. For the horseshoe prior hierarchical form specified in [Eq. 3](#), the density $\kappa_j | \lambda, \tau^{-1}, \tilde{v}_i$ is given as

$$p(\kappa_j | \lambda, \tau^{-1}, \tilde{v}_i) = \frac{1}{\pi} \frac{k_j}{(k_j^2 - 1) \kappa_j + 1} \frac{1}{\sqrt{\kappa_j(1 - \kappa_j)}}, \quad \kappa_j \in (0, 1),$$

$$\text{where } k_j = \lambda \sqrt{\sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1} \xi^2 \tilde{v}_i}}.$$

[Carvalho et al. \(2010\)](#) and [Polson and Scott \(2010\)](#) have demonstrated a close connection between the horseshoe shrinkage weights $(1 - \kappa_j)$ as “pseudo-posterior” probabilities and posterior inclusion probabilities (PIP) from the two-group mixture models such as spike-and-slab priors. They also suggest selecting all features with $1 - \kappa_j > 0.5$, although [Piironen and Vehtari \(2017\)](#) have found that such a criterion would fail in the case of a large number of correlated features. Nevertheless, a cutoff other than 0.5 may be adopted and it is still reasonable to consider the cutoff for determining selected features as a judgment call under the one-group Horseshoe priors. On the other hand, with two-group spike-and-slab priors, a median probability threshold of 0.5 is commonly used to compare with the PIPs for selecting features ([Barbieri and Berger, 2004](#); [Barbieri et al., 2021](#); [Fan et al., 2024](#)), and other adaptive thresholds can be determined using Bayesian FDRs ([Muller et al., 2007](#)). The difference in interpreting and applying the cutoffs to determine selected features lies in distinct mechanisms of imposing shrinkage between one-group and two-group sparse priors.

The auxiliary mixture sampling techniques developed in the context of mean field variational Bayes ([Wand et al., 2011](#)) turn out to be crucial for setting up efficient Gibbs samplers under normal mean models ([Makalic and Schmidt, 2015](#)). Specifically, via an

auxiliary variable v_j , the half-Cauchy distribution that the local shrinkage parameter s_j follows can be expressed as a scale mixture of inverse-Gamma distributions:

$$\begin{aligned} s_j^2 | v_j &\sim \text{Inverse-Gamma}(1/2, 1/v_j), \quad j = 1, \dots, p, \\ v_j &\sim \text{Inverse-Gamma}(1/2, 1). \end{aligned} \quad (4)$$

A similar hierarchical form for the global shrinkage parameter λ can also be obtained. Combined, the robust Bayesian horseshoe prior regression can be hierarchically formulated as:

$$\begin{aligned} y_i | \beta_0, \mathbf{x}_i, \boldsymbol{\beta}, \tilde{v}_i &\sim N(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^{-1} \xi^2 \tilde{v}_i), \\ \beta_0 &\sim N(0, \sigma_{\beta_0}^2), \\ \tilde{v}_i | \tau &\sim \text{Exp}(\tau^{-1}), \\ \tau &\sim \text{Gamma}(e, f), \\ \beta_j | s_j, \lambda &\sim N(0, \lambda^2 s_j^2), \\ s_j^2 | v_j &\sim \text{Inverse-Gamma}(1/2, 1/v_j), \\ \lambda^2 | \xi_1 &\sim \text{Inverse-Gamma}(1/2, 1/\xi_1), \\ v_j &\sim \text{Inverse-Gamma}(1/2, 1), \\ \xi_1 &\sim \text{Inverse-Gamma}(1/2, 1). \end{aligned} \quad (5)$$

2.2.2. Robust horseshoe + prior regression

The horseshoe + prior proposed by Bhadra et al. (2017) has the following hierarchical form:

$$\begin{aligned} s_j &\sim C^+(0, \phi_j), \\ \phi_j &\sim C^+(0, 1), \\ \lambda &\sim C^+(0, 1), \end{aligned} \quad (6)$$

where independent half-Cauchy priors centered at zero with an half-Cauchy mixing variable ϕ_j have been assigned on the local shrinkage hyperparameters $(s_1, \dots, s_p)$. Compared with the original horseshoe hierarchical formulation in (3), incorporating ϕ_j leads to an extra level of local shrinkage that makes s_j 's conditionally independent given ϕ_j .

The horseshoe + estimator is a refinement of the original horseshoe estimator, tailored for ultra-sparse signals. Bhadra et al. (2017) have shown that horseshoe + achieves superior performance by reducing the posterior mean squared error and accelerating posterior concentration under the Kullback–Leibler divergence. The extra level of hierarchy in (6) induces a shrinkage prior with a sharper spike at zero, thus resulting in more aggressive shrinkage on noises near zero. Meanwhile, the nested Cauchy priors produce a heavier tail compared to the horseshoe prior, reducing shrinkage on larger signals. Together, it facilitates better recovery of strong signals and greater separation of signals and noise in ultra-sparse models. This is in spirit very close to the “theoretically ideal” and “gold standard” spike-and-slab priors (Bai et al., 2021; Ročková, 2018; Carvalho et al., 2010) but has a computational advantage due to its one-group nature.

Marginalizing over ϕ_j results in the following density for s_j :

$$p(s_j) = \frac{4}{\pi^2} \frac{\log(s_j)}{s_j^2 - 1}.$$

Propositions 1 and 2 on the shrinkage factor κ_j under robust horseshoe regression models naturally extend to the horseshoe + setting. Since the full conditional distribution of β_j remains unchanged, Proposition 1 holds for horseshoe + prior as well. Derivations of the following proposition are available in the Appendix.

Proposition 3. For the horseshoe + prior formulated in (6), the density $\kappa_j | \lambda, \tau^{-1}, \tilde{v}_i$ is given by

$$p(\kappa_j | \lambda, \tau^{-1}, \tilde{v}_i) = \frac{2}{\pi^2} \frac{\log\left(\sqrt{\frac{1-\kappa_j}{\kappa_j k_j^2}}\right)}{1 - \kappa_j (1 + k_j^2)} \frac{k_j}{\sqrt{\kappa_j (1 - \kappa_j)}}, \quad \kappa_j \in (0, 1).$$

where $k_j = \lambda \sqrt{\sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1} \xi^2 \tilde{v}_i}}.$

Again, by following the auxiliary mixture sampling (Wand et al., 2011) and resulting Gibbs sampling under sparse normal mean models (Makalic and Schmidt, 2015), we can express the robust Bayesian hierarchical model with horseshoe + prior as:

$$\begin{aligned}
 y_i \mid \beta_0, \mathbf{x}_i, \beta, \tilde{v}_i &\sim N(\beta_0 + \mathbf{x}_i^\top \beta, \tau^{-1} \xi^2 \tilde{v}_i), \\
 \beta_0 &\sim N(0, \sigma_{\beta_0}^2), \\
 \tilde{v}_i \mid \tau &\sim \text{Exp}(\tau^{-1}), \\
 \tau &\sim \text{Gamma}(e, f), \\
 \beta_j \mid s_j, \lambda &\sim N(0, \lambda^2 s_j^2), \\
 s_j^2 \mid v_j &\sim \text{Inverse-Gamma}(1/2, 1/v_j), \\
 v_j \mid \phi_j^2 &\sim \text{Inverse-Gamma}(1/2, 1/\phi_j^2), \\
 \phi_j^2 \mid \zeta_j &\sim \text{Inverse-Gamma}(1/2, 1/\zeta_j), \\
 \zeta_j &\sim \text{Inverse-Gamma}(1/2, 1), \\
 \lambda^2 \mid \xi_1 &\sim \text{Inverse-Gamma}(1/2, 1/\xi_1), \\
 \xi_1 &\sim \text{Inverse-Gamma}(1/2, 1).
 \end{aligned}$$

2.2.3. Robust Bayesian regularized horseshoe regression

The hierarchical models for Horseshoe and Horseshoe + prior regression depend on extremely heavy tails of the half-Cauchy prior, which has infinite variance. Therefore, regression parameters with very large magnitude are not regularized at all and can still be sampled, leading to potential numeric instability in MCMC. In addition, there is no principled way to incorporate sparsity belief in specifying the global shrinkage parameters λ . To overcome the limitations, Piironen and Vehtari (2017) have proposed to regularize the Horseshoe prior as follows:

$$\begin{aligned}
 \beta_j \mid s_j, \lambda, b &\sim N\left(0, \lambda^2 \tilde{s}_j^2\right), \quad \tilde{s}_j^2 = \frac{b^2 s_j^2}{b^2 + \lambda^2 s_j^2}, \\
 s_j &\sim C^+(0, 1), \quad j = 1, \dots, p,
 \end{aligned} \tag{7}$$

where $\tilde{s}_j^2$ is a modified local shrinkage parameter with a positive tuning parameter b . Under robust Horseshoe prior regression, Proposition 1 indicates that the relative amount of shrinkage can be expressed as $\kappa_j = \left(1 + \lambda^2 s_j^2 a_j\right)^{-1}$. Correspondingly, the shrinkage factor for robust regression with regularized Horseshoe prior is $\tilde{\kappa}_j = \left(1 + \lambda^2 \tilde{s}_j^2 a_j\right)^{-1}$. Therefore, if $b \rightarrow \infty$, regularized Horseshoe induces the same amount of shrinkage imposed by Horseshoe. On the other hand, $\tilde{\kappa}_j = 1$ as $b \rightarrow 0$, so the regression coefficient is not penalized. When $0 < b < \infty$, regularized Horseshoe performs a more aggressive shrinkage estimation as $\tilde{\kappa}_j > \kappa_j$.

In fact, regularized Horseshoe has a conceptual connection to Bayesian elastic net (Li and Lin, 2010), although it has not been explicitly mentioned in Piironen and Vehtari (2017). Recall that a Bayesian elastic net prior is given as:

$$\pi(\beta_j \mid \lambda_1, \lambda_2) \propto \exp\left(-\lambda_1 |\beta_j| - \lambda_2 \beta_j^2\right),$$

where λ_1 and λ_2 are positive tuning parameters associated with Laplacian and Gaussian priors, respectively. The hierarchical structure of regularized Horseshoe in (7) can be viewed as replacing the Laplace prior using a Horseshoe prior while keeping the Gaussian kernel. Consequently, a regularized horseshoe prior can also be expressed as:

$$\pi(\beta_j \mid s_j, \lambda, b) \propto N(0, \lambda^2 s_j^2) N(0, b^2) \propto N(0, \lambda^2 \tilde{s}_j^2),$$

where a Gaussian prior $N(0, b^2)$ on β_j is multiplied to the horseshoe prior in order to mimic the role that a ridge penalty plays in elastic net (Zou and Hastie, 2005). Therefore, regularized Horseshoe is an extension of the original horseshoe prior to account for structured sparsity in terms of multicollinearity. Zou and Hastie (2005) have provided a detailed rationale on why incorporating ridge term in penalized variable selection can accommodate correlations among predictors, which has partially justified the advantage of analyzing correlated genomics features using regularized Horseshoe demonstrated in Piironen and Vehtari (2017). The connection between regularized horseshoe prior and Bayesian elastic net is also important for addressing the central question in our study, that is, how essential is exact sparsity for facilitating accurate robust Bayesian inference in high-dimensional regression. We will demonstrate more details in numerical studies of this paper.

It has been recommended to specify either a fixed value or a prior distribution for b . In the absence of strong prior knowledge about the scale of the relevant coefficients, we generally advise assigning a prior to b rather than fixing it. A common and practical choice is to place an inverse-gamma prior on b^2 , such that $b^2 \sim \text{Inv-Gamma}(\frac{c}{2}, \frac{d}{2})$ (Piironen and Vehtari, 2017).

Our limited literature mining indicates that the hierarchical model for robust Bayesian regression with regularized horseshoe priors has not been established yet. We specify the hierarchical model below:

$$\begin{aligned}
 y_i &| \beta_0, \mathbf{x}_i, \beta, \tilde{v}_i \sim N(\beta_0 + \mathbf{x}_i^\top \beta, \tau^{-1} \xi^2 \tilde{v}_i), \\
 \beta_0 &\sim N(0, \sigma_{\beta_0}^2), \\
 \tilde{v}_i | \tau &\sim \text{Exp}(\tau^{-1}), \\
 \tau &\sim \text{Gamma}(e, f), \\
 \beta_j &| s_j, \lambda, b \sim N(0, \lambda^2 s_j^2) N(0, b^2), \\
 s_j^2 &| v_j \sim \text{Inverse-Gamma}(1/2, 1/v_j), \\
 \lambda^2 &| \xi_1 \sim \text{Inverse-Gamma}(1/2, 1/\xi_1), \\
 v_j &\sim \text{Inverse-Gamma}(1/2, 1), \\
 \xi_1 &\sim \text{Inverse-Gamma}(1/2, 1), \\
 b^2 &\sim \text{Inverse-Gamma}(c/2, d/2).
 \end{aligned}$$

2.3. Summary of all methods under comparison

We summarize the Gibbs sampling procedures for the robust Bayesian regression models with the horseshoe prior (RBHS), the horseshoe + prior (RBHS +), and the regularized horseshoe prior (RBRHS) introduced in Section 2.2 in Table 2. We have also provided a concise summary of the proposed RBHS, RBHS +, and RBRHS, together with their non-robust counterparts BHS, BHS +, and BRHS (Piironen and Vehtari, 2017; Makalic and Schmidt, 2015) in Table 3 below. Detailed Gibbs samplers for posterior inferences for all six methods under comparison are shown in Section D of the Appendix.

3. Simulation

We perform a thorough evaluation of the robust Bayesian regression with horseshoe, horseshoe +, and regularized horseshoe priors, which are termed as RBHS, RBHS + and RBRHS, respectively. Their non-robust alternatives are BHS, BHS + and BRHS. Please refer to Section 2.3 for a detailed summary. The datasets are simulated based on model (1) under two configurations: one with sample size $n = 200$ and dimensionality $p = 600$, and the other with a larger scale of $n = 400$ and $p = 600$. The predictors are drawn from a multivariate normal distribution with (1) an autoregressive (AR(1)) correlation structure where the i th and j th predictors have correlation $0.5^{|i-j|}$, and (2) a banded correlation structure where the i th and j th predictors have correlation 0.5 if $|i - j| = 1$, and zero otherwise. The true regression coefficient vector is defined as $\beta = (\beta_1, \beta_2, \dots, \beta_p)^\top$, where 15 components are nonzero and independently drawn from a Uniform(0.4, 0.9) distribution, and the remaining entries are set to zero. The intercept is fixed at one, i.e., $\beta_0 = 1$. We consider five error distributions in simulation: $N(0, 1)$ (Error 1), $t(2)$ (Error 2), Laplace(0, 1) (Error 3); 80% $N(0, 1) + 20\%$ $N(0, 3)$ (Error 4), and Lognormal(0, 1) (Error 5). We additionally examine the scenario with *non-i.i.d.* random errors by adopting the following data-generating model:

$$y_i = 1 + \mathbf{x}_i^\top \beta + (1 + x_{i2})\epsilon_i, \quad \text{for } i = 1, \dots, n,$$

The *i.i.d.* error terms ϵ_i in model (1) are replaced with heteroscedastic errors of the form $(1 + x_{i2})\epsilon_i$, while the regression coefficients remain unchanged from the *i.i.d.* error setting.

The simulation study is performed over 100 independent replicates. For each of the six Bayesian methods, posterior samples were obtained from 10,000 MCMC iterations, following the removal of the initial 5000 iterations used as burn-in. For variable selection with the horseshoe family of models, although Carvalho et al. (2010) have suggested selecting all features with $1 - \kappa_j > 0.5$, where $(1 - \kappa_j)$ represents horseshoe shrinkage weights, Piironen and Vehtari (2017) have demonstrated the failure of this criterion in the presence of multicollinearity. Therefore, rather than using κ_j , we follow Bhadra et al. (2017) and van der Pas et al. (2017) among others to adopt the 95% credible interval for variable selection. Specifically, a predictor is considered selected if its 95% credible interval excludes zero. An additional benefit of this approach is that the same 95% credible intervals are used to assess the validity of our inference procedures, ensuring that variable selection and inference are conducted under a consistent and coherent criterion. The identified features are subsequently used to calculate the number of true positives (TPs), false positives (FPs), F1 score (F1) and Matthews correlation coefficient (MCC). Specifically, the F1 score is defined as $F1 = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}$, and the MCC is defined as $\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}$, where TN and FN refer to true negatives and false negatives, respectively. We use the posterior medians $\hat{\beta}$ to compute the L_1 estimation error as $\sum_{j=1}^p |\beta_j - \hat{\beta}_j|$.

Variable selection and estimation Table 4 shows the identification and estimation results under the AR(1) correlation structure with $(n, p) = (200, 600)$ across all the five error distributions. The three robust Bayesian procedures, RBRHS, RBHS +, and RBHS, consistently demonstrate superior performance over their non-robust counterparts (BRHS, BHS +, and BHS), especially under heavy-tailed and skewed errors. For instance, under the $t(2)$ error (Error 2), all three robust methods demonstrate much better performance than their non-robust counterparts. RBHS + achieves the highest identification accuracy with TPs of 8.182 (sd 3.332) and FPs of 0.009 (sd 0.095), followed by RBHS with TPs of 7.659 (sd 3.368) and FPs of 0.012 (sd 0.109), and RBRHS with TPs of 5.878 (sd

Table 2

Gibbs samplers for RBHS, RBHS+, and RBRHS.

Algorithm 1: Gibbs Samplers for RBHS, RBHS+, and RBRHS1. **Initialize:** Set initial values of $\beta_j, \beta_0, \tilde{v}_i, \tau$ for one of the models from RBHS, RBHS+, and RBRHS:

- RBHS: $s_j^2, \nu_j, \lambda^2, \xi_1$.
- RBHS+: $s_j^2, \nu_j, \phi_j^2, \zeta_j, \lambda^2, \xi_1$.
- RBRHS: $s_j^2, \nu_j, \lambda^2, \xi_1, b^2$.

2. Set iteration counter $t = 0$.3. **While** $t < 10000$ **do:**

Update parameters for one of the models (RBHS, RBHS+, or RBRHS) chosen from Step 1.

(a) $\beta_0 \sim N(\mu_{\beta_0}, \sigma_{\beta_0}^2)$. Sampling the intercept.(b) **RBHS:**

- $\beta_j \sim N(\mu_{\beta_j}, \sigma_{\beta_j}^2)$, where

$$\sigma_{\beta_j}^{-2} = \sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1}\xi^2\tilde{v}_i} + \frac{1}{\lambda^2 s_j^2}, \quad \mu_{\beta_j} = \sigma_{\beta_j}^2 \sum_{i=1}^n \frac{x_{ij}(y_i - \beta_0 - \sum_{k \neq j} x_{ik}\beta_k)}{\tau^{-1}\xi^2\tilde{v}_i}. \quad \text{Sampling } \beta_j.$$
- $\nu_j \sim \text{Inverse-Gamma}\left(1, 1 + \frac{1}{s_j^2}\right)$. Sampling ν_j .

(c) **RBHS+:**

- $\beta_j \sim N(\mu_{\beta_j}, \sigma_{\beta_j}^2)$, where

$$\sigma_{\beta_j}^{-2} = \sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1}\xi^2\tilde{v}_i} + \frac{1}{\lambda^2 s_j^2}, \quad \mu_{\beta_j} = \sigma_{\beta_j}^2 \sum_{i=1}^n \frac{x_{ij}(y_i - \beta_0 - \sum_{k \neq j} x_{ik}\beta_k)}{\tau^{-1}\xi^2\tilde{v}_i}. \quad \text{Sampling } \beta_j.$$
- $\nu_j \sim \text{Inverse-Gamma}\left(1, \frac{1}{s_j^2} + \frac{1}{\phi_j^2}\right)$. Sampling ν_j .
- $\phi_j^2 \sim \text{Inverse-Gamma}\left(1, \frac{1}{\nu_j} + \frac{1}{\zeta_j}\right)$. Sampling ϕ_j^2 .
- $\zeta_j \sim \text{Inverse-Gamma}\left(1, \frac{1}{\phi_j^2} + 1\right)$. Sampling ζ_j .

(d) **RBRHS:**

- $\beta_j \sim N(\mu_{\beta_j}, \sigma_{\beta_j}^2)$, where

$$\sigma_{\beta_j}^{-2} = \sum_{i=1}^n \frac{x_{ij}^2}{\tau^{-1}\xi^2\tilde{v}_i} + \left(\frac{1}{\lambda^2 s_j^2} + \frac{1}{b^2}\right), \quad \mu_{\beta_j} = \sigma_{\beta_j}^2 \sum_{i=1}^n \frac{x_{ij}(y_i - \beta_0 - \sum_{k \neq j} x_{ik}\beta_k)}{\tau^{-1}\xi^2\tilde{v}_i}. \quad \text{Sampling } \beta_j.$$
- $\nu_j \sim \text{Inverse-Gamma}\left(1, 1 + \frac{1}{s_j^2}\right)$. Sampling ν_j .
- $b^2 \sim \text{Inverse-Gamma}\left(\frac{c+p}{2}, \frac{d + \sum_{j=1}^p \beta_j^2}{2}\right)$. Sampling b^2 .

(e) $\tilde{v}_i \sim \text{Inverse-Gaussian}\left(\sqrt{\frac{2\xi^2}{(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2}}, 2\tau\right)$. Sampling $\tilde{v}_i$.(f) $\tau \sim \text{Gamma}\left(e + \frac{3n}{2}, f + \sum_{i=1}^n \tilde{v}_i + \sum_{i=1}^n \frac{(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\xi^2\tilde{v}_i}\right)$. Sampling τ .(g) $s_j^2 \sim \text{Inverse-Gamma}\left(1, \frac{\beta_j^2}{2\lambda^2} + \frac{1}{\nu_j}\right)$. Sampling s_j^2 .(h) $\lambda^2 \sim \text{Inverse-Gamma}\left(\frac{p+1}{2}, \frac{1}{\xi_1} + \frac{1}{2} \sum_{j=1}^p \frac{\beta_j^2}{s_j^2}\right)$. Sampling λ^2 .(i) $\xi_1 \sim \text{Inverse-Gamma}\left(1, 1 + \frac{1}{\lambda^2}\right)$. Sampling ξ_1 .4. Increment $t = t + 1$.**Output:** Posterior samples for all model parameters under RBHS, RBHS+, and RBRHS.**Table 3**

Summary of all methods under comparison.

Methods	Names	Robust	Likelihood	Variable Selection	Reference
RBHS	Robust Bayesian regression with horseshoe prior	Yes	Laplacian	Horseshoe prior	newly proposed
RBHS+	Robust Bayesian regression with horseshoe+ prior	Yes	Laplacian	Horseshoe+ prior	newly proposed
RBRHS	Robust Bayesian regression with regularized horseshoe prior	Yes	Laplacian	Regularized horseshoe prior	newly proposed
BHS	Bayesian regression with horseshoe prior	No	Gaussian	Horseshoe prior	Carvalho et al. (2010)
BHS+	Bayesian regression with horseshoe+ prior	No	Gaussian	Horseshoe+ prior	Bhadra et al. (2017)
BRHS	Bayesian regression with regularized horseshoe prior	No	Gaussian	Regularized horseshoe prior	Piironen and Vehtari (2017)

3.763) and FPs of 0.010 (sd 0.100). In contrast, the non-robust methods show inferior performance: BHS yields TPs of 3.390 (sd 3.128) with FPs of 0.159 (sd 2.234); BHS+ gives TPs of 3.350 (sd 3.144) with FPs of 0.182 (sd 2.860); and BRHS reports TPs of 3.074 (sd 3.037) with FPs of 0.100 (sd 1.906). In terms of F1 score and MCC, which provide a more comprehensive assessment of variable selection accuracy, the robust methods also outperform their non-robust counterparts. For example, under the $t(2)$ error (Error 2), RBHS+ leads to the largest F1 score of 0.676 (sd 0.208) and an MCC of 0.722 (sd 0.160), while RBHS has a slightly smaller F1 score of 0.644 (sd 0.215) and an MCC of 0.696 (sd 0.167). RBRHS yields an F1 score of 0.515 (sd 0.270) and an MCC of 0.614 (sd 0.196). In comparison, the non-robust approaches result in lower metric values. Specifically, BHS attains an F1 score of 0.324 (sd 0.256) and an MCC of 0.489 (sd 0.187). BHS+ shows an F1 score of 0.320 (sd 0.258) and an MCC of 0.487 (sd 0.190), while BRHS records an F1 score of 0.297 (sd 0.253) and an MCC of 0.477 (sd 0.184). In terms of L1 error, Table 4 clearly indicates that robust methods generally lead to smaller estimation errors.

In addition to reporting numerical results in Table 4, we further present boxplots of the F1 score, MCC, and L_1 estimation error from Table 4 in Fig. 1. The visualization provides a complementary summary of method performance under five *i.i.d.* model errors. Fig. 1 displays the results for *i.i.d.* errors. Under $N(0,1)$ error (Error 1), all six methods perform well, with F1 and MCC scores tightly clustered near 1, and small estimation errors. As the error distributions become heavy-tailed or skewed (Errors 2-5), we observe a clear difference in performance. The robust RBHS, RBHS+, and RBRHS all maintain high identification accuracy (F1 and MCC) and low estimation error, with RBHS+ generally achieving the narrowest interquartile ranges (IQR) and lowest medians across all metrics. In contrast, non-robust BHS, BHS+, and BRHS demonstrate increased variability and deteriorated performance, particularly under the contaminated normal (Error 4) and log-normal error (Error 5) settings. These trends indicate the competitive performance of robust regression with horseshoe family of priors to outliers and heavy-tailed errors.

In total, there are eight settings under different sample size ($n=200$ or 400 , $p=600$), correlation structure (AR(1) vs. banded), and error types (*i.i.d.* vs. non-*i.i.d.*). Except Table 4, identification and estimation performance for the other seven settings is listed in Tables S1 – S7 in the appendix, respectively. Specifically, Tables S1, S2, and S3 correspond to $n=200$, with Tables S1 and S2 focusing on banded correlation under *i.i.d.* and non-*i.i.d.* errors, respectively, and Table S3 reporting AR(1) with non-*i.i.d.* errors. Tables S4 – S7 extend these scenarios to $n=400$, with Tables S4 and S5 addressing *i.i.d.* errors under AR(1) and banded structures, and Tables S6 and S7 covering non-*i.i.d.* errors. Combined, these tables demonstrate the superior variable selection and estimation performance of three robust Bayesian approaches (RBHS, RBHS+, and RBRHS) under heteroscedastic model errors. Besides, in the Appendix, we choose to visualize Table S2 using Figure S1, which clearly illustrates that the robust methods are competitive.

Remark on distinct identification and prediction patterns Table 4 in the main text and Tables S1 – S7 in the appendix suggests that, in general, the number of false positives is very low, often close to zero, which is sometimes achieved at the cost of identifying significantly fewer true positives. For example, under the $t(2)$ error in Table 4, the number of false positives for all three robust horseshoe models is nearly zero, but their true positives are also very low, which is only about half of the total number of true signals. This phenomenon suggests that horseshoe prior regressions aggressively shrink regression coefficients toward zero, even if exact sparsity is not naturally accommodated.

Under sparse normal mean models, Bhadra et al. (2017) have theoretically established that horseshoe+ priors lead to better prediction performance in terms of posterior mean squared error compared to horseshoe priors. Table 4 shows that RBHS+ consistently results in the smallest L_1 estimation errors. Prediction and estimation performance are typically related, although they are two different aspects assessing model fitting. Our results partially justify the superior prediction performance of horseshoe+ priors with the corresponding theoretical results established in Bhadra et al. (2017).

High-dimensional inference for robust Bayesian horseshoe regression We have also evaluated the six methods based on their ability to quantify uncertainty. The data have been generated from model(1) under two settings: $(n, p) = (100, 500)$ and $(n, p) = (200, 1000)$. The predictors are generated from a multivariate normal distribution with an autoregressive (AR(1)) correlation structure where the i th and j th predictors have correlation $0.5^{|i-j|}$. The true regression coefficient vector is $\beta = (1, 1.5, 2, 0, \dots, 0)^T$, where the first three entries are nonzero ($\beta_1 = (1, 1.5, 2)^T$) and the remaining coefficients are zero ($\beta_2 = (0, \dots, 0)^T$). The intercept is set to zero. Two error distributions are considered for ϵ : a standard normal distribution $N(0,1)$ and a heavy-tailed $t(2)$ distribution. The simulation study is based on 1000 replicates. We note that this data generating model has been adopted to assess the validity of inference for two-group spike-and-slab priors (Song and Liang, 2023; Fan et al., 2024). Therefore, high-dimensional inference results from different studies can be directly compared.

Table 5 shows 95% empirical coverage probabilities for all the six methods under $N(0,1)$ and $t(2)$ model errors. With $N(0,1)$ error, RBHS, RBHS+, and RBRHS yield coverage probabilities for $\beta_1 (=1)$, $\beta_2 (=1.5)$ and $\beta_3 (=2)$ ranging from 0.933 to 0.940, all close to the 0.95 nominal level, except that RBRHS leads to a low coverage probability 0.853 for β_3 . We observe competitive performance in inference for the non-robust BHS, BHS+ and BRHS as the empirical coverage probabilities are slightly below 0.95, except that RBHS provides an under-coverage of 0.825 for β_3 with its marginal credible intervals. We also observe that the lengths of the credible intervals across all six methods are similar, fluctuating around 0.5. For the zero coefficients β_2 , Table 5 shows a super efficient phenomenon with 100% coverage probability and much shorter interval length, which is consistent with the observation made for spike-and-slab priors under the same data generating model (Song and Liang, 2023; Fan et al., 2024).

The pattern of inference results under heavy-tailed $t(2)$ error is even more interesting. We now observe that robust methods distinguish from non-robust counterparts with more accurate coverage probabilities and narrower marginal credible intervals. Although heavy-tailed model errors lead to wider credible intervals compared to those under $N(0,1)$ error for non-zero coefficients, RBHS, RBHS+ and RBRHS generally yield empirical coverage probabilities around 0.95, with interval lengths below 1. In contrast, the interval lengths for the non-robust BHS, BHS+, and BRHS are all much greater than 1, and their coverage probabilities deviate from

Table 4

Identification and estimation results of RBHS, RBHS +, RBRHS, BHS, BHS + and BRHS for the datasets with AR(1) correlation under *i.i.d.* errors, $n = 200$ and $p = 600$. Mean(sd) of true positives (TP), false positives (FP), F1 score (F1), MCC and L1 error based on 100 replicates.

	Methods	TP	FP	F1	MCC	L1 error
Error 1 N(0,1)	RBHS	14.663(0.618)	0.059(0.260)	0.986(0.024)	0.986(0.024)	1.887(0.424)
	RBHS +	14.665(0.611)	0.079(0.298)	0.986(0.024)	0.986(0.024)	1.456(0.376)
	RBRHS	14.639(0.625)	0.050(0.240)	0.986(0.023)	0.986(0.023)	1.863(0.394)
	BHS	14.698(0.575)	0.039(0.209)	0.988(0.022)	0.988(0.022)	1.745(0.409)
	BHS +	14.713(0.552)	0.051(0.246)	0.988(0.021)	0.988(0.021)	1.324(0.349)
	BRHS	14.660(0.596)	0.023(0.150)	0.987(0.022)	0.987(0.022)	1.724(0.372)
Error 2 $t(2)$	RBHS	7.659(3.368)	0.012(0.109)	0.644(0.215)	0.696(0.167)	3.832(1.507)
	RBHS +	8.182(3.330)	0.009(0.095)	0.676(0.208)	0.722(0.160)	3.345(1.545)
	RBRHS	5.878(3.763)	0.010(0.100)	0.515(0.270)	0.614(0.196)	4.766(1.910)
	BHS	3.390(3.128)	0.159(2.234)	0.324(0.256)	0.489(0.187)	8.633(33.848)
	BHS +	3.350(3.144)	0.182(2.860)	0.320(0.258)	0.487(0.190)	8.123(35.855)
	BRHS	3.074(3.037)	0.100(1.906)	0.297(0.253)	0.477(0.184)	8.083(28.971)
Error 3 Laplace(0,1)	RBHS	12.818(1.742)	0.030(0.187)	0.916(0.073)	0.919(0.067)	2.286(0.721)
	RBHS +	13.021(1.671)	0.034(0.221)	0.924(0.070)	0.927(0.065)	1.797(0.644)
	RBRHS	12.578(1.918)	0.011(0.130)	0.906(0.082)	0.911(0.075)	2.339(0.734)
	BHS	11.982(1.987)	0.071(0.286)	0.880(0.089)	0.886(0.080)	3.017(0.853)
	BHS +	12.009(2.041)	0.061(0.256)	0.881(0.092)	0.887(0.082)	2.352(0.778)
	BRHS	11.709(2.095)	0.040(0.211)	0.868(0.096)	0.876(0.085)	2.907(0.790)
Error 4 80%N(0,1) + 20%N(0,3)	RBHS	11.679(2.144)	0.027(0.162)	0.867(0.099)	0.875(0.087)	2.577(0.809)
	RBHS +	11.978(2.065)	0.039(0.209)	0.880(0.095)	0.886(0.085)	2.093(0.729)
	RBRHS	11.124(2.462)	0.018(0.133)	0.840(0.121)	0.852(0.104)	2.746(0.860)
	BHS	10.012(2.488)	0.085(0.319)	0.785(0.130)	0.804(0.108)	3.835(1.051)
	BHS +	10.021(2.514)	0.071(0.293)	0.786(0.132)	0.805(0.110)	3.102(1.012)
	BRHS	9.629(2.585)	0.046(0.232)	0.766(0.139)	0.788(0.115)	3.671(10.992)
Error 5 Lognormal(0,1)	RBHS	11.843(2.739)	0.005(0.071)	0.869(0.133)	0.879(0.115)	2.058(0.990)
	RBHS +	12.251(2.538)	0.006(0.077)	0.888(0.120)	0.896(0.105)	1.700(0.891)
	RBRHS	11.102(3.415)	0.007(0.095)	0.827(0.186)	0.848(0.149)	2.464(1.311)
	BHS	6.290(3.722)	0.076(0.290)	0.545(0.257)	0.624(0.195)	5.632(1.963)
	BHS +	6.257(3.749)	0.060(0.269)	0.542(0.260)	0.621(0.200)	4.980(2.039)
	BRHS	5.859(3.731)	0.056(0.259)	0.514(0.265)	0.603(0.200)	5.479(1.928)

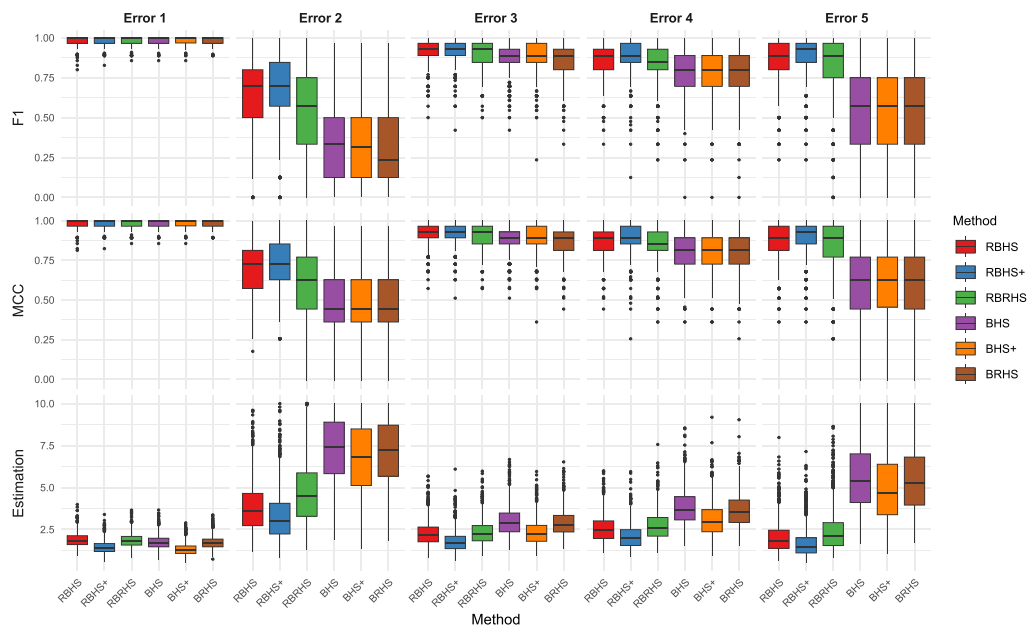


Fig. 1. F1 score, MCC score and Estimation error under AR(1) correlation, *i.i.d.* error and $(n, p) = (200, 600)$ corresponding to Table 4.

the nominal level, mostly falling in the range of 0.7 to 0.8. Similar to the pattern under $N(0,1)$ error, RBHS and RBHS+ maintain stable performance under heavy-tailed errors, RBRHS shows a notable lower coverage probability of 0.805 for β_3 .

The behavior of RBRHS under heavy-tailed model errors indicates a key challenge in high-dimensional inference: balancing satisfactory empirical coverage probability with the optimal interval length. Table 5 suggests that RBRHS struggles to achieve nominal coverage probabilities for some of non-zero coefficients while maintaining the narrowest interval length, compared to RBHS and RBHS+. We conjecture that this is due to the different mechanisms of constructing these priors. As shown in Section 2.2.3, regularized Horseshoe is tightly connected to Bayesian elastic net that accounts for structured sparsity in terms of multicollinearity (Li and Lin, 2010). In high-dimensional regression, performing accurate statistical inference becomes increasingly challenging when the model involves complex structures beyond predictor independence. Bayesian elastic net itself does not yield valid marginal credible intervals. In contrast, both the Horseshoe and Horseshoe+ priors have simpler structures that do not explicitly account for correlations or other forms of structured sparsity. Therefore, both lead to more accurate inference results under robust models.

A further comparison among interval lengths shows consistency between the numerical results and prior theoretical results established in literature. Table 5 shows that RBHS+ typically yields shorter Bayesian credible intervals compared to RBHS, which can be explained by more aggressive shrinkage induced by construction of the Horseshoe+ prior. Note that RBRHS introduces additional amount of shrinkage through the Gaussian prior that mimics the ridge penalty in elastic net, so the interval lengths are also smaller compared to RBHS. The pattern also suggests that the shrinkage imposed through the Gaussian prior under RBRHS is even more extreme than the amount due to the additional layer of half-Cauchy prior on local shrinkage parameters under RBHS+. Besides, across all Bayesian approaches, the super-efficiency phenomenon persists for noise variables: the coverage probabilities for the zero coefficients remain at 1.000.

Fig. 2 and Fig. 3 provide visual summaries of the inference results reported in Table 5, corresponding to the $N(0,1)$ and $t(2)$ error settings, respectively. For illustration, we display results for the first 10 predictors, where the first 3 have nonzero coefficients and the remaining 7 are truly zero. The true coefficient values are indicated by black horizontal bars. Each plot overlays the intervals obtained from 1000 replicates, with black segments denoting intervals that include the true value and red segments indicating non-coverage. The empirical coverage probability for each coefficient is shown above its corresponding bar. Figs. 2 and 3 clearly highlight the inferential advantage of using robust Bayesian sparse regression with the horseshoe family of priors.

We also evaluate the inference performance under a much larger scale setting with $(n, p) = (200, 1000)$, and present detailed results in Figures S2 and S3 and Table S8 in the appendix. We observe a similar pattern, where the robust Bayesian methods with the family of horseshoe priors continue to outperform their non-robust counterparts in terms of coverage and size of the interval. In particular, a larger sample size leads to slightly better empirical coverage probabilities and narrower marginal credible intervals under non-zero coefficients, reflecting improved stability and more accurate posterior inference. These results in supplementary demonstrate the utility of the proposed approaches for inference in both moderate- and high-dimensional settings.

Additional Evaluations We assess MCMC convergence using the potential scale reduction factor (PSRF) (Gelman and Rubin, 1992; Brooks and Gelman, 1998), which indicates convergence when values are close to 1. In line with the recommendation by Gelman et al. (1995), we adopt 1.1 as the convergence threshold. In our analysis, the PSRF values for all nonzero parameters fall below this benchmark after discarding burn-in samples, confirming that the chains have successfully converged to the stationary distribution. Figures S4 to S6 in the appendix illustrate the convergence behavior of the robust methods under *i.i.d.* Error 2 with $n = 200$ and $p = 600$. To further demonstrate the advantage of our methods over published robust Bayesian variable selection methods including horseshoe prior quantile regression (Kohns and Szendrei, 2024), we have performed comparison studies as shown in Table S9 and Table S10 in the appendix. In the appendix, Table S11 summarizes computational time and implementations, and Table S12 reports effective sample sizes for the robust Bayesian methods. In a more extreme high-dimensional setting, Table S13 presents variable-selection and estimation results for the proposed methods as well as benchmarks such as spike-and-slab models and frequentist regularization methods. These results further establish the advantage of proposed methods.

4. Real data analysis

We apply all the methods under comparison to analyze an expression quantitative trait locus (eQTL) dataset generated using 120 twelve-week-old male offspring of laboratory rats for a study to analyze RNA from the eyes of these animals based on microarrays containing over 31,042 probe sets (Scheetz et al., 2006). Following Huang et al. (2008) and Wang et al. (2012a), we have performed two pre-processing steps to filter out genes that are not expressed or lack variations. First, we have excluded any probe whose maximum expression across the 120 rats falls below the 25th percentile of all expression values; and second, we have excluded any probe whose expression range across the 120 rats is less than 2. The preprocessing step leads to a total of 3057 probes. We are interested in detecting gene expressions that are associated with gene TRIM32 (represented by probe 1389163_at) which has been reported to cause retinal disorders in humans. As noted by Scheetz et al. (2006), any genetic element demonstrated to influence the expression of a particular gene or gene family implicated in a specific disease serves as a strong candidate for involvement in that disease—either as a primary contributor or as a genetic modifier. We rank all remaining probes based on their coefficient of variation (CV) and select the top 300 probes with the highest CV values. These selected probes are subsequently kept for downstream analysis.

Specifically, RBHS, RBHS+, and RBRHS, along with their non-robust counterparts, BHS, BHS+, and BRHS, have been adopted for analysis. The 95% Bayesian credible intervals have been used to determine whether probes are selected or not. We present the number of selected probes and their pairwise overlaps of all the six methods in Table 6. The diagonal entries of Table 6 are the total number of nonzero coefficients identified by each method, while the off-diagonal entries represent the number of overlapping probes selected by each pair of methods. Since all the six methods belong to the horseshoe family, it is not surprise to observe that their

Table 5
95% empirical coverage probabilities assessed over 1000 replications for RBHS, RBHS +, RBRHS, BHS, BHS + and BRHS using the datasets with AR(1) covariates; $n = 100$ and $p = 500$.

		Methods					
		RBHS	RBHS +	RBRHS	BHS	BHS +	BRHS
N(0,1)	Coverage of β_1						
	β_1	0.936	0.937	0.939	0.934	0.937	0.941
	β_2	0.940	0.938	0.940	0.943	0.943	0.957
	β_3	0.940	0.933	0.853	0.942	0.944	0.825
	Average length						
	β_1	0.500	0.496	0.472	0.491	0.486	0.460
	β_2	0.552	0.550	0.515	0.546	0.542	0.503
	β_3	0.492	0.489	0.470	0.487	0.484	0.464
	Coverage of β_2	1.000	1.000	1.000	1.000	1.000	1.000
	Average length	0.101	0.100	0.090	0.273	0.094	0.076
t(2)	Coverage of β_1						
	β_1	0.926	0.930	0.934	0.779	0.773	0.799
	β_2	0.957	0.956	0.965	0.847	0.848	0.886
	β_3	0.969	0.967	0.805	0.892	0.903	0.892
	Average length						
	β_1	0.917	0.891	0.793	1.230	1.211	1.166
	β_2	0.972	0.943	0.784	1.506	1.491	1.400
	β_3	0.808	0.795	0.720	1.328	1.321	1.251
	Coverage of β_2	1.000	1.000	1.000	1.000	1.000	1.000
	Average length	0.101	0.080	0.083	0.273	0.210	0.206

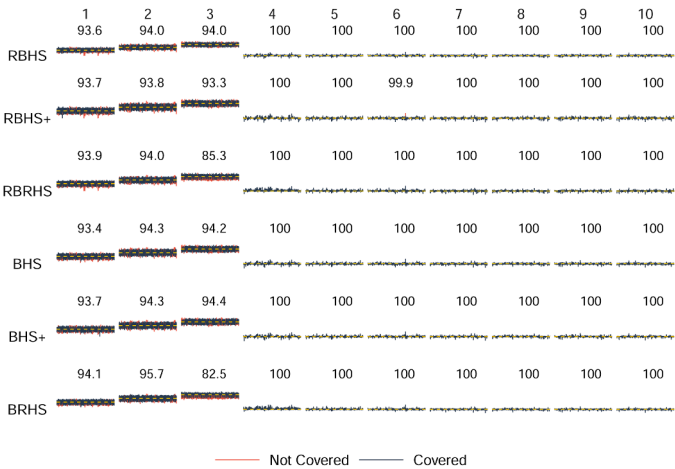


Fig. 2. Credible intervals with empirical coverage probabilities under a 95% nominal level are shown for the first 10 predictors, with the first 3 having true nonzero coefficients. Results are based on 1,000 simulated datasets under a homogeneous model with AR(1) covariate correlation, $(n, p) = (100, 500)$, and N(0,1) error.

Table 6
Variable selection results for the case study. The numbers of probes identified by different Bayesian methods and their overlaps.

	RBHS	RBHS +	RBRHS	BHS	BHS +	BRHS
RBHS	18	9	6	11	3	5
RBHS +		23	5	10	3	2
RBRHS			25	9	7	9
BHS				32	4	6
BHS +					13	3
BRHS						15

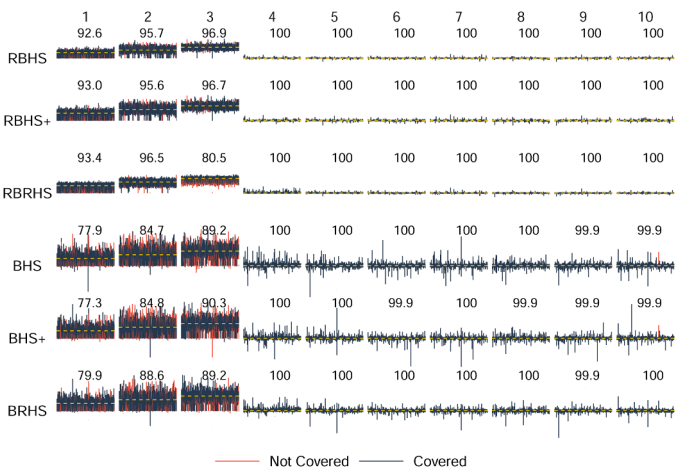


Fig. 3. Credible intervals with empirical coverage probabilities under a 95% nominal level are shown for the first 10 predictors, with the first 3 having true nonzero coefficients. Results are based on 1,000 simulated datasets under a homogeneous model with AR(1) covariate correlation, $(n, p) = (100, 500)$, and $t(2)$ error.

Table 7
Multi-split analysis of the microarray dataset.

Method	Model size	Random partition	
		Average model size	Prediction error
RBHS	18	16.620(6.227)	0.098(0.013)
RBHS +	23	15.860(6.719)	0.099(0.013)
RBRHS	25	26.320(7.113)	0.097(0.014)
BHS	32	52.580(36.331)	0.790(2.239)
BHS +	13	23.020(14.098)	0.115(0.023)
BRHS	15	49.280(34.450)	0.485(0.817)

findings show certain degree of overlap. In addition, some findings are method-specific, reflecting differences in variable selection sensitivity between robust and non-robust models.

We have also assessed the prediction performance of all the six methods through multi-splitting of the microarray data. Specifically, we perform 50 random data splits, each consisting of 80 randomly selected rats for training and the remaining 40 rats for testing. For each of the 50 random partitions, we fit models on the training data and evaluate predictive accuracy on the held-out test set using mean absolute deviation (MAD), defined as $MAD = \frac{1}{40} \sum_{i=1}^{40} |y_i - \hat{y}_i|$. Table 7 lists the average number of nonzero regression coefficients (Ave # nonzero) obtained across 50 random data partitions, along with the corresponding average MADs. The standard errors for both summary statistics are also provided. Overall, the results show that the robust methods (RBHS, RBHS +, RBRHS) achieve much lower prediction errors compared to the non-robust methods (BHS, BHS +, BRHS), while also maintaining a more parsimonious model with smaller model size. For example, RBHS achieves a prediction error of 0.098 (sd 0.013) with 16.620 (sd 6.227) predictors. On the other hand, the non-robust BHS yields a prediction error of 0.790 (sd 2.239) with 52.580 (sd 36.331) predictors. BHS results in a substantially larger number of selected predictors and greater variance in model size, further demonstrating heterogeneity of the data and benefit of robust Bayesian high-dimensional variable selection.

As with any variable selection method, different repetitions may identify different subsets of key predictors. In Table S14 from the appendix, the columns list the probes selected using the complete dataset, along with how frequently these probes appeared in the final models across 50 random partitions for the six methods: RBHS, RBHS +, RBRHS, BHS, BHS +, and BRHS. The probes are arranged in order of decreasing frequency within each method. From Table S14, we observe that certain probes, such as 1368778_at, are consistently selected with high frequency across both robust (RBHS, RBHS +, RBRHS) and non-robust (BHS, BRHS) methods, indicating their overall selection stability. In contrast, probes such as 1370539_at, 1370810_at, 1370056_at, 1370359_at, and 1370946_at show high selection frequencies exclusively under robust method but are rarely or never selected by non-robust approaches, reflecting the unique sensitivity of robust frameworks in uncovering signals that may be masked by outliers or model misspecification. This suggests that while some key predictors are universally robust and consistently detected across methods, others are uniquely captured by robust modeling, offering complementary insights and highlighting the added value of robust approaches in high-dimensional variable selection.

5. Discussion

In this study, we have robustified the existing horseshoe family of (horseshoe, horseshoe+, and regularized horseshoe) priors by employing a robust likelihood based on the Laplace distribution. Because the Laplace distribution corresponds to the asymmetric Laplace distribution at the 50th quantile level, the proposed methods naturally lend themselves to potential extensions toward Bayesian quantile horseshoe regression and further comparison with the horseshoe prior quantile regression (Kohns and Szendrei, 2024). However, extensions to quantile regression framework is not trivial, as the well known quantile crossing problem, or the occurrence of non-monotone fitted quantile functions, needs to be carefully addressed. This issue becomes more pressing when estimating extreme quantiles in high-dimensional settings and when shrinkage priors induce heterogeneous sparsity across different quantile levels. Developing Bayesian quantile regression with the horseshoe family of priors that mitigate quantile crossing while maintaining valid statistical inference represents an important direction for our future work.

We have comprehensively assessed the performance of robust Bayesian sparse regression with the horseshoe family of priors. Our numeric results indicate that the proposed RBHS, RBHS+ and RBRHS are superior in terms of variable selection, shrinkage estimation and statistical inference compared with their non-robust alternatives in the presence of heavy-tailed model errors. In particular, RBHS and RBHS+ demonstrate competitive performance in yielding marginal credible intervals with nominal coverage probabilities in high-dimensions. These advantages make the robust horseshoe family appealing for applied use; however, our results also indicate some limitations. Specifically, the regularized versions (RBRHS and BRHS) cannot maintain valid inference, since by construction they accommodate strong correlations among predictors. Similar phenomena have also been observed in frequentist and Bayesian regularization approaches, such as elastic net methods that incorporate structured sparsity, where addressing multicollinearity makes valid inference substantially more challenging. Thus, while robust horseshoe priors perform well in standard sparse settings, further investigation and development of horseshoe based methods that can deliver reliable inference under data structures more closely mimicking real cancer genomics applications is warranted.

Our study has shown that even though horseshoe family of priors cannot lead to exact zero parameter estimates, or exact sparsity, it can still maintain valid inference under sparse linear models. While this interesting phenomenon is seemingly not consistent with the previous findings that exact sparsity facilitates valid inference in high-dimensional scenarios (Fan et al., 2024), a closer examination of the numerical study suggests otherwise. Table 4 and tables alike demonstrate that the number of false features identified by all the six methods are very close to 0, sometimes at the cost of reducing true positive findings for a uniform shrinkage on all regression coefficients, suggesting that horseshoe family of priors tends to encourage sparse findings even exact 0 posterior estimates are not obtained.

A similar observation in inference can be made in penalized generalized estimation equation (PGEE) for longitudinal data in frequentist studies (Wang et al., 2012b). Although the Newton-Raphson model fitting procedure cannot yield exact zero estimates for the PGEE estimator, Wang et al. (2012b) have established asymptotic distribution of PGEE estimator and validated the inference procedure on finite samples. Nevertheless, beyond the linear models, longitudinal studies show that lacking exact sparsity causes issues in estimation by treating group selection as sparse-group selection (Zhou et al., 2019, 2022). Therefore, robust Bayesian estimation and inference with the horseshoe family of priors worth further exploration under more complicated sparse structures.

Statistical inference for robust Bayesian variable selection methods has been largely overlooked in the literature. van der Pas et al. (2017) is among the few works that investigate uncertainty quantification for the horseshoe prior, but the results remain limited to low-dimensional settings with Gaussian errors. Our study focuses on applied and computational aspects and does not provide theoretical justification on why valid inference results can be observed under long-tailed errors. We conjecture that such results may be established by extending the adaptive posterior concentration results developed for the horseshoe priors (van der Pas et al., 2017) to high-dimensional models with heavy-tailed model errors.

Declaration of competing interest

The authors declare that there is no conflict of interest.

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Supplementary material

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